

C226

Research Design and Analysis

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

Two schools in a district that just hired a new superintendent test a math program. All students at School A use the program; School B continues as usual. Students were not randomly assigned. What design is this?

- A. Quasi-experimental (nonequivalent groups)
- B. Correlational
- C. Case study
- D. True experiment

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

What single change would turn that two-school study into a true experiment?

- A. Double the sample, since large samples always make a design experimental.
- B. Randomly select the two schools from across the state.
- C. Add a pretest.
- D. Randomly assign students (or classes) to the program or the usual curriculum.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

At a school known for its popular garden club, all first graders use a new reading program from September to May. Scores rise sharply between the fall and spring tests. There is no comparison group. What is the main threat to concluding the program caused the gain?

- A. No threat; the gains prove the program works.
- B. Selection bias between groups
- C. Regression to the mean
- D. Maturation

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A researcher at an online university with students in all 50 states compares mean final-exam scores across three course formats: live online, recorded, and hybrid. Which test fits?

- A. Chi-square test
- B. One-way ANOVA
- C. Pearson correlation
- D. Independent-samples t-test

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A researcher asks whether preference for online vs. in-person classes is associated with student level (undergraduate vs. graduate). Which test fits?

- A. Pearson correlation
- B. Chi-square test of independence
- C. Independent-samples t-test
- D. One-way ANOVA

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A research team that includes two retired principals sets $\alpha = .05$ and finds $p = .03$ for the difference between a program group and a comparison group. What is the correct interpretation?

- A. Reject the null; if there were truly no difference, results this extreme would occur about 3% of the time.
- B. The study proves the program works.
- C. The program has a large effect.
- D. There is a 3% chance the null hypothesis is true.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A study finds $p = .21$ at $\alpha = .05$, and the researcher concludes the program has no effect. In reality, the program does work. What happened?

- A.** A Type I error
- B.** Only random sampling error, which is not a decision error.
- C.** A Type II error
- D.** No error, because $p > .05$ proves the null.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A district data team using a new dashboard finds $r = -0.60$ between daily screen time and GPA among 2,000 high school students. Which conclusion is warranted?

- A.** Students with more screen time tend to have lower GPAs, but this does not show screen time causes lower GPAs.
- B.** The relationship is weak because r is negative.
- C.** Screen time causes lower GPAs.
- D.** Screen time explains 60% of the variation in GPA.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A national study of 40,000 students, funded by a federal grant, finds a statistically significant difference ($p < .001$): mean scores of 501 vs. 500, with a standard deviation of 100. What is the best interpretation?

- A.** A large, important effect because $p < .001$.
- B.** Statistically significant but practically trivial ($d = 0.01$).
- C.** Not a real result, since significance with large samples is always an error.
- D.** A large effect, $d = 1.0$.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Two groups differ by 6 points, and Cohen's $d = 0.5$. What is the standard deviation?

- A.** 6
- B.** About 0.08
- C.** 3
- D.** 12

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A randomized experiment at one small suburban private school shows large gains from a writing program designed by a former journalist. The superintendent of a large urban district wants to adopt it right away. What is the main caution?

- A.** The study had no control group.
- B.** None; random assignment guarantees results hold in every school.
- C.** External validity: results from one small private school may not generalize to a large urban district.
- D.** Internal validity is weak because students were randomly assigned.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A researcher interviewing 12 teachers from a school network that meets monthly has coded transcript segments as “no planning period,” “too many meetings,” and “grading at home.” What is the next analytic step?

- A.** Stop; the codes are the findings.
- B.** Delete any code that appears only once, since only frequent codes matter.
- C.** Group related codes into categories and develop a theme such as time scarcity, checking it against the data.
- D.** Run a t-test on how often each code appears.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

Two schools in a district ~~that just hired a new superintendent~~ test a math program. All students at School A use the program; School B continues as usual. Students were not randomly assigned. What design is this?

Target word: classify the design **Decides:** Intact groups, no random assignment.

A	Correct	A treatment is compared across groups that were not randomized.
B	Adjacent category	A treatment was introduced and compared; that is not a correlational design.
C	Confusion set	A case study examines one bounded case in depth.
D	Dropped element	Drops the missing random assignment.

2. Answer D Concept 1 • transfer

What single change would turn that two-school study into a true experiment?

Target word: the defining feature **Decides:** Random assignment to conditions.

A	Absolute language	Size does not substitute for random assignment.
B	Confusion set	Random sampling aids generalization; it does not create equivalent groups.
C	Adjacent category	Strengthens a quasi-experiment but does not make it true.
D	Correct	Random assignment defines a true experiment.

3. Answer D Concept 2

At a school ~~known for its popular garden club~~, all first graders use a new reading program from September to May. Scores rise sharply between the fall and spring tests. There is no comparison group. What is the main threat to concluding the program caused the gain?

Target word: internal-validity threat **Decides:** Young children grow a lot over a school year anyway.

A	Absolute language	Without a comparison, other explanations remain.
B	Adjacent category	There is only one group.
C	Confusion set	That needs selection on extreme scores; all first graders took part.
D	Correct	Natural development over nine months could explain the gains.

4. Answer B Concept 3

A researcher at an online university ~~with students in all 50 states~~ compares mean final-exam scores across three course formats: live online, recorded, and hybrid. Which test fits?

Target word: choose the test **Decides:** Three group means on a continuous outcome.

A	Confusion set	For categorical outcomes, not mean scores.
B	Correct	Compares means across three or more groups.
C	Adjacent category	Relates two continuous variables; format is categorical.
D	Dropped element	Handles only two groups; drops the third.

5. Answer B Concept 3 • transfer

A researcher asks whether preference for online vs. in-person classes is associated with student level (undergraduate vs. graduate). Which test fits?

Target word: test for two categorical variables **Decides:** Both variables are categories.

A	Dropped element	Drops that neither variable is continuous.
B	Correct	Tests association between categorical variables.
C	Adjacent category	Needs a continuous outcome.
D	Confusion set	Needs a continuous outcome and compares means.

6. Answer A Concept 4

A research team ~~that includes two retired principals~~ sets $\alpha = .05$ and finds $p = .03$ for the difference between a program group and a comparison group. What is the correct interpretation?

Target word: meaning of p **Decides:** $p < \alpha$.

A	Correct	That is the definition of p.
B	Absolute language	Statistical tests support, not prove.
C	Adjacent category	p says nothing about effect size.
D	Confusion set	p is not the probability the null is true.

7. Answer C Concept 4 • transfer

A study finds $p = .21$ at $\alpha = .05$, and the researcher concludes the program has no effect. In reality, the program does work. What happened?

Target word: name the error **Decides:** Failed to reject a false null.

A	Inverted operation	Type I rejects a true null, the opposite.
B	Adjacent category	The wrong conclusion is itself a Type II decision error.
C	Correct	Missing a real effect.
D	Absolute language	Failing to reject is not proof of no effect.

8. Answer A Concept 5

A district data team ~~using a new dashboard~~ finds $r = -0.60$ between daily screen time and GPA among 2,000 high school students. Which conclusion is warranted?

Target word: interpret r **Decides:** Observational correlation.

A	Correct	Moderate-to-strong negative association, no causal claim.
B	Confusion set	Sign shows direction; magnitude shows strength.
C	Adjacent category	Correlation cannot establish cause.
D	Wrong stem number	Shared variance is $r^2 = 36\%$.

9. Answer B Concept 6

A national study of 40,000 students, ~~funded by a federal grant~~, finds a statistically significant difference ($p < .001$): mean scores of 501 vs. 500, with a standard deviation of 100. What is the best interpretation?

Target word: practical importance **Decides:** $d = (501 - 500) / 100 = 0.01$.

A	Adjacent category	p is not effect size.
B	Correct	A huge sample makes a tiny difference significant.
C	Absolute language	The difference is real, just tiny.
D	Dropped element	Uses the raw 1-point difference and drops the division by SD.

10. Answer D Concept 6 • transfer

Two groups differ by 6 points, and Cohen's $d = 0.5$. What is the standard deviation?

Target word: solve for SD **Decides:** $SD = 6 / 0.5$.

A	Wrong stem number	The mean difference, not the SD.
B	Confusion set	Divides d by the difference.
C	Inverted operation	Multiplies 6×0.5 .
D	Correct	$6 / 0.5 = 12$.

11. Answer C Concept 7

A randomized experiment at one small suburban private school shows large gains from a writing program ~~designed by a former journalist~~. The superintendent of a large urban district wants to adopt it right away. What is the main caution?

Target word: limit on applying results **Decides:** One very different setting.

A	Dropped element	A randomized experiment has one; misreads the design.
B	Absolute language	Assignment is not sampling from all schools.
C	Correct	Strong internal validity does not ensure transfer to new settings.
D	Inverted operation	Random assignment strengthens internal validity.

12. Answer C Concept 8

A researcher interviewing 12 teachers ~~from a school network that meets monthly~~ has coded transcript segments as "no planning period," "too many meetings," and "grading at home." What is the next analytic step?

Target word: next step after coding **Decides:** Codes are grouped into categories and themes.

A	Dropped element	Drops the move from codes to themes.
B	Absolute language	Rare codes can matter and may disconfirm a theme.
C	Correct	Codes → categories → themes.
D	Adjacent category	Applies a quantitative test to interpretive analysis.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	10	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	3	
Absolute language	Always, never, guarantee — where judgment applies.	7	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	True vs. quasi-experiment	Random assignment of participants to conditions makes a true experiment; comparing intact, non-randomized groups is quasi-experimental.	Confusing random sampling with random assignment.	Existing classes or schools compared with no random assignment.
2	Threats to internal validity	Without a comparison group, change can come from maturation, history, testing, or regression to the mean rather than the treatment.	Treating pre-post gains as proof the program worked.	One group measured before and after a long program.
3	Choosing a statistical test	Two group means → t-test; three or more means → ANOVA; two categorical variables → chi-square; two continuous variables → correlation.	Picking a test without checking the number of groups and the type of data.	Count the groups and name the level of measurement.
4	p-values and error types	p is the probability of data this extreme if the null were true; rejecting a true null is Type I, failing to reject a false null is Type II.	Reading p as the probability the null is true, or as effect size.	A p-value compared to alpha, or a wrong conclusion to name.
5	Correlation is not causation	r gives direction and strength of a linear relationship; r^2 is shared variance; neither shows cause.	Inferring cause, or reading a negative r as weak.	An r value from observational data.
6	Effect size vs. significance	Cohen's d = mean difference / SD; with huge samples, trivial differences can be statistically significant.	Equating a small p with a large or important effect.	Very large N with a tiny mean difference.
7	External validity	Strong internal validity does not guarantee results generalize to other populations, settings, or times.	Thinking random assignment makes findings apply everywhere.	A result from one setting proposed for a very different one.
8	Qualitative analysis	Qualitative analysis moves from codes to categories to themes, checked back against the data.	Stopping at codes or treating code counts as the analysis.	Labeled transcript segments and "next step."

Course scope (public description): The Research Design and Analysis course focuses on applying strategies for effective design of empirical research studies. Particular emphasis is placed on selecting or constructing the design that will provide the most valid results, analyzing the kind of data that would be obtained, and making defensible interpretations and drawing appropriate conclusions based on the data. Research Questions and Literature Review is a prerequisite for this course.